



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST



MASS FLOW GAUGE

MODEL DF-231BA

ERROR AND TROUBLESHOOTING MANUAL

1. ERRORS

If an error occurs, the digital indicator will show an error number, such as E-01, and the number remains shown there until the power switch is turned off. Check the contents of the error by referring to the following table.

ERR signals are output in the following cases.

Error type	Description	Cause	Error Code
Instrument error	EEP ROM error	EEP ROM malfunction	E-01
	System error	CPU ROM operation faulty	E-02
	RAM error	RAM operation faulty	E-03
Sensor error	Analog input range over *1	Sensor faulty or internal circuit malfunction	E-04
	Sensor output error	Sensor faulty	—
External signal input operation error	Zero adjustment (ZIN) operation error	Zero adjustment is made when there is a deviation of over ± 50 count (3.5-digit display set) or ± 500 count (4.5-digit display set)	—

*1 **E-04** occurs in case sensor output exceeds the full-scale when power is turned on. The measurement mode recovers as the sensor output returns to the range within the full-scale.

2. TROUBLESHOOTING

Check the following points before ordering to have your device repaired.

Symptom	Cause	Countermeasures
No power is supplied.	No LEDs are lit on.	Check that the power cord was correctly connected.
No key operation is accepted.	When a key is pressed, the alarm beeps.	Check that the keyboard lock was not set. In the setup mode, set Loc or oFF.
	When the Set key is pressed after something is set in the setup mode, the alarm beeps, and the setting cannot be validated.	Check that Loc in setup mode was not set to on1 or on2. If Loc was set to on1 or on2, set it to oFF.
	The Auto Zero key is invalid.	- Check that this function was not set through the external input. (External input has a higher priority than direct keyboard operation.) - Check that Loc in setup mode was not set to on1. - If Loc is on1, set it to on2 or oFF.
No external input (I/O) is accepted.	Any of the external input signals are invalid.	- Check that the connectors were correctly connected. - Check the input voltage of pin 37 (EXTPSIN) of the external I/O connector (DATA I/O). - In 5V input specification, if a voltage higher than 5V is applied to pin 37 (EXTPSIN) of the external I/O connector, the internal circuits may be damaged.
No input from the RS-232C is accepted.	No input from the RS-232C is valid. No response returns even if a command is entered.	- Check that the connectors were correctly connected. - Check that the signal specifications of the host computer are correct. - Check that the baud rate of the host computer agrees with that of the DF-231BA. - Check that the main unit is not in the setup mode. (No measurement is effected in the setup mode, and so no response returns at all.)
	Error (error field: 80) returns.	- Check that the correct command was used. - Check that the command was entered in upper case. (Commands should be entered in upper case).
	Error (error field: 10) returns.	Check that this function was not set through the external input. (External input has a higher priority than RS-232C.)
	Signal is transmitted although there must be no flow.	Check for any leak of the fluid from the tube.
There is a deviation from the reference flow.		- Check for any leak of the fluid from the tube. - Check for any foreign matter, such as dirt or oil, in the tube or connecting port. - Check whether the flow is considerably unstable or the flow is substantially out of the measurement range.